Spectral-Spatial Genetic Algorithm-Based Unsupervised Band Selection for Hyperspectral Image Classification

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***Abstract*— Band selection (BS) can mitigate the "curse of dimensionality" problem and improve the performance of hyperspectral image (HSI) classification. Genetic algorithms (GAs) have been applied to the task of hyperspectral BS showing significant advantages compared with other literature methods. However, the traditional GAs-based methods often select sets of bands having residual redundancy due to the large search space related to hyperspectral BS and the limitation of premature convergence in GAs. Moreover, existing GAs-based methods often are supervised, and that needs a large number of labeled samples to compute the fitness value for assessing the quality of selected bands. In this article, an unsupervised BS approach based on an improved GA is proposed. A fitness function based on the fisher score combined with superpixel is designed for evaluating the discriminability of band subsets considering both spectral and spatial information. Then, modified genetic operations are constructed to restrain the search space and reduce the redundancy of selected bands. The performance of the proposed spectral-spatial GA-based BS method is evaluated on three HSIs. The experimental results demonstrate that the proposed method is superior to the traditional GA-based method and seven state-of-the-art unsupervised methods.**

***Index Terms*— Band selection (BS), fisher score, genetic algorithm (GA), hyperspectral image (HIS), remote sensing superpixels.**

I. Introduction

HYPERSPECTRAL sensors can acquire images with tens or even hundreds of narrow and contiguous spectral bands in a wide range from visible to thermal infrared wavelengths [1].

Indeed, classification is one of the most active areas of research in the field of hyperspectral imagery (HSI) analysis [2]. The high spectral resolution of HSI offers rich spectral information representing the details of ground objects. However, the resulting numerous spectral channels have high redundancy and strong correlation that affect the performance of HSI classification [3]. In such a situation, a large number of labeled samples are required to guarantee a satisfactory classification accuracy for avoiding the "curse of dimensionality" (also known as "Hughes phenomenon") [4], [5]. Meanwhile, many hyperspectral systems acquire images that also have a high spatial resolution. This may reduce statistical separation among different land-cover classes in the spectral domain for the increase of intraclass variability and the decrease of interclass variability [6]. In this context, the collection of a sufficient number of training samples becomes difficult and impractical in many real applications [7].

In order to improve classification performance in the HSI classification task, considering the limited number of labeled samples, it is crucial to be able to reduce the number of channels of HSI given as input to the classifier. Dimensionality reduction techniques presented in the literature can be generally divided into two main categories: feature extraction (FE) and band selection (BS), also known as feature selection (FS) [8]. The purpose of FE is to learn a linear or nonlinear transformation to transform the space of original spectral bands into a new feature subspace with lower dimension and higher classification potential [9]. BS aims to select an appropriate subset of bands from the original spectral channels according to a given specific selection criterion [10]. Compared with FE methods, BS can retain the original spectral information of HSI, which is beneficial to the interpretation of selected bands subset in the subsequent analysis.

According to the use of prior information, BS techniques are classified into three types: supervised, semisupervised, and unsupervised methods. Supervised BS methods [11]–[13] usually reach high classification accuracies due to their training on labeled samples. However, the robustness of supervised BS methods is greatly influenced by the representativeness of training samples. Semisupervised BS methods [14]–[16] take advantage of a small number of labeled samples that are integrated with a large amount of unlabeled data to select

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bands. Unsupervised BS methods [17]–[19] determine the BS without any training sample. In this article, we focus on the unsupervised BS technique.

In unsupervised methods, two BS schemes, i.e., ranking and clustering, are often used by considering the amount of information contained in bands or their degree of correlation (or both).In the ranking-based methods, such as information divergence (ID) [20] and the maximum-variance principal component analysis (MVPCA) [21], the significance of each band is assessed by a given criterion by computing the commonly used statistical characteristics, e.g., non-Gaussianity (NG) [22] and variance. Then, bands are selected according to their scores. However, ranking-based methods often ignore the spectral correlation among bands, which result in selecting a subset of bands containing redundant information. Clustering-based methods partition bands into several groups according to a defined similarity metric. Then, the centers of clusters are taken as representative of the bands to form the final subset of bands. Recently, some effective clustering algorithms, such as affinity propagation (AP) [23], [24] and a fast density peak-based clustering algorithm (FDPC) [25], have introduced in the BS task, resulting in the adaptive AP (AAP) [26], the superpixel metric learning-based AP (SML-AP), and the superpixel representation learning-based AP (SRL-AP) [27]. Moreover, a few ranking-based clustering methods, such as the enhanced FDPC (E-FDPC) [28], were proposed. The purpose of the above BS methods is to select highly discriminative and low-redundant band subsets. However, in ranking-based and clustering-based schemes, each band is evaluated independently and then combined with others to form the subset of bands. Thus, the discriminative ability of the selected subset of bands is not considered during the BS process.

From the perspective of the search of selected bands, BS can be considered as a typical combinatorial optimization problem. Evolutionary algorithms (EAs) (e.g., genetic algorithms (GAs) [29], simulated annealing algorithms [30], [31], particle swarm optimization (PSO) [32], [33], firefly algorithm [34], and colony optimization [35]) are population-based metaheuristic optimization algorithms that have been used as solutions to search HSI BS iteratively. EA-based methods usually encode the bands into a random binary space. Then, operations inspired by the behavior of the biological system or the laws of natural phenomena are adopted to search the optimal subset of bands. To measure the quality of the selected bands, a predefined evaluation criterion, i.e., a fitness function, is used during the search process. GAs, as the most representative EAs, can get satisfactory solutions in various optimization problems. GAs are naturally suitable for solving the task of BS for the HSI. Ghamisi and Benediktsson [36] integrated the GA with PSO for BS. Gong *et al.* [17] proposed a multiobjective optimization BS method by using an information entropy-based evaluation criterion for finding the optimal tradeoff result. In [37], a low-order approximation of the maximum joint mutual information criterion is introduced to the clonal selection algorithm for BS. In [38], a GA coupled with the spectral angle mapper (SAM) is presented to select the subset of bands for Earth compositional mapping. However, existing GA-based methods are based on quality metrics that require a large number of labeled samples. Moreover, the bands selected by traditional GA-based methods may have non-negligible redundancy due to the huge search space for hyperspectral BS and the limitations of premature convergence in GAs.

HSI represents real land surfaces that typically extend for a few pixels in the image. Thus, nearby pixels in HSI have high probabilities of belonging to the same class, i.e., there is a high spatial correlation. The spatial information defined as spatial homogeneity has been exploited to improve the performance of BS. There are two main types of spatial neighborhood systems used to describe the spatial relationship of pixels in HSI, i.e., the rectangular neighborhood (e.g., pixel-centered local or global fixed neighborhood) [13], [16], [39]–[41] and the irregular adaptive neighborhood. Obviously, the rectangular neighborhood cannot reflect the complex spatial structure of HSI. Superpixel [42] refers to a coherent and local grouping of pixels in regions. For HSI, a superpixel is designed as a ground object block consisting of adjacent pixels with similar texture and spectral information. It can preserve most of the spatial structure information and align better with object edges than fixed image patches. Moreover, it can reduce the complexity of subsequent image analysis tasks [43], [44]. Superpixels can be generated by over-segmentation of an image using a superpixel segmentation algorithm [45]–[47]. Therefore, the pixels included in superpixels describe an irregular neighborhood that is adaptively adjusted according to the characteristics of the pixels. Superpixels have achieved remarkable successes in HSI classification [48]–[50], endmember detection [51], target detection [52], [53], and image decomposition [54]. They have also been considered for the BS in HSI [27].

In this article, an unsupervised GA-based BS approach based on the superpixel concept is proposed for selecting an effective subset of bands for classification of HSI. The main objective of the proposed approach is to address the BS problem under the critical conditions of large spectral variability of HSI and taking into account the difficulty in collecting labeled samples of many real applications. This approach aims at improving the BS process from the following two aspects. First, a spectral-spatial fitness function combined with superpixel for the GA is designed. It is based on a fisher score criterion that evaluates the discrimination ability of a series of potential band subsets (referred to as individuals), taking into account the spectral separability and spatial homogeneity during the evolutionary process. Second, new genetic operations are developed to restrain the search space for guaranteeing that the potential band subsets are represented by individuals containing a small fixed number of low redundancy bands during the BS process. Accordingly, the proposed approach is named spectral-spatial GA-based unsupervised BS (SSGA). Experimental results obtained on three HSI data sets show that the proposed approach can provide robust highly discriminative and low-redundant bands subset.

The rest of this article is organized as follows. In Section II, the proposed SSGA approach is presented in detail, including the evolutionary criterion construction, the design of chromosome coding, and the definition of fitness function and genetic operations with superpixels. Section III reports the

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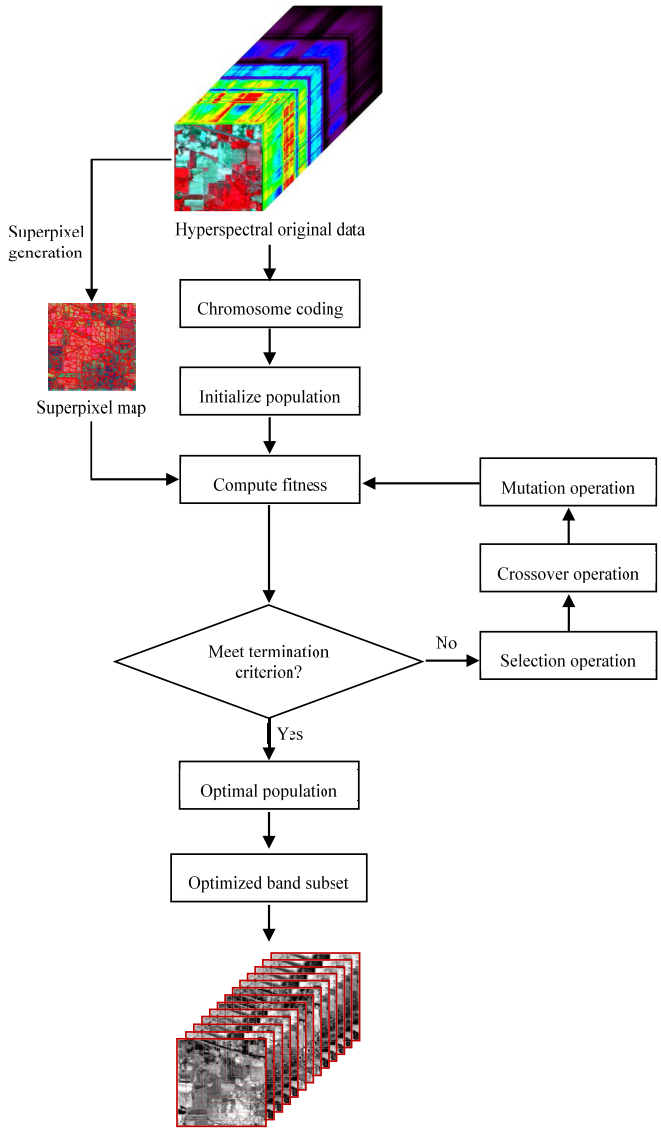


Fig. 1. Flowchart of the proposed SSGA BS approach.

experimental results of the proposed SSGA and reference methods and discusses the effectiveness and the stability of the considered BS methods. Finally, conclusions are drawn in Section IV.

II. Proposed SSGA BS Method

In this section, an SSGA-based BS is proposed for HSI classification. The flowchart of the proposed SSGA approach is illustrated in Fig. 1. In the SSGA approach, we first introduce the chromosome coding and fitness function applied to the BS criterion with superpixel. Then, we design the genetic operations (i.e., selection, crossover, mutation) to determine a desired number of bands to be selected.

Given an HSI data set, $X = \{x_1, x_2, \ldots, x_L\} \subset \mathbb{R}^{N \times L}$, where $x_i = \{x_{i1}, x_{i2}, \ldots, x_{xiN}\}$, N is the number of pixel vectors present in the HSI, L is the number of spectral bands, and x_i represents the ith vector of spectral responses at a pixel location in the HSI. BS task can be modeled as a special combinatorial optimization problem aimed to select the representative subset of bands $Y = \{y_1, y_2, \ldots, y_l\}$, $y_l \in X$, $(l \ll L)$.

A. Chromosome Coding

In traditional chromosome coding, nearly half of the genes should be involved in the initialization process. For decreasing this randomness, we present an improved chromosome coding mode. In this mode, a chromosome expresses a candidate result of HSI BS, and each initialized chromosome only includes a small fixed number of bands. Given a chromosome with L genes, each gene represents a spectral band and controls the corresponding band being absent or present in the considered situation. Thus, the value of each gene can be either 0 or 1. A chromosome is initialized being subject to l in a binary space; l is the desired number of selected bands. Then, a set of potential solutions (i.e., chromosomes) with low redundancy can be established.

B. Fitness Function

The fitness function is used to evaluate the quality of solutions, which is the quantitative evaluation of the quality of the chromosome. Whether a chromosome should be selected in the successive generation depends on its fitness value. The fitness function directly affects the convergence speed of the GA and whether the optimal solution can be found. Most GA-based BS methods take classification accuracy as the fitness value that relies on labeled samples. However, land-cover class labels are difficult and expensive to obtain in many HSI applications. Moreover, labeled samples are mainly concentrated on the spectral information but do not reflect the complex spatial structure of HSI. In this article, we design a spectral-spatial fitness function by defining a superpixel-based unsupervised trace ratio criterion to measure the discrimination ability of potential band subsets with respect to the spectral separability and spatial homogeneity of the HSI.

At present, many superpixel segmentation algorithms have been proposed. They can be generally divided into two categories: gradient descent-based methods and graph theory-based methods. The basic idea of the former is to start from a coarse clustering to modify the clustering results iteratively through the gradient descent method until the condition of convergence is satisfied. Then, the superpixels can be obtained by transforming superpixel segmentation into an energy function minimization. The whole image is considered as an undirected weighted graph, in which a vertex denotes the corresponding pixel in the image, the adjacent relation of pixels can be regarded as the edge between corresponding vertexes, and the weight of an edge is given by the similarity or difference between the corresponding pixels. Finally, a variety of segmentation criteria are adopted on the segmentation of vertexes in the graph to generate superpixels.

Entropy rate superpixel (ERS) [45] is one of the representative methods for superpixel extraction. It is a graph theory-based method. The ERS has shown distinct advantages in preprocessing or postprocessing of HSI. In ERS, a new objective function is constructed consisting of two components, i.e., an entropy rate of the random walk on a graph term and

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a balancing term. The entropy rate ensures the compactness and homogeneity of superpixels. The balancing term restrains the number of superpixels with an unbalanced size. Compared with the other algorithms, the superpixels segmented with ERS exhibit high homogeneity and edge overlapping with the real ground objects. In this article, the ERS is adopted to construct homogeneous regions, i.e., superpixels in the definition of unsupervised trace ratio criterion. Let $SP = \{sp_1, sp_2, \ldots, sp_K\}$ denote the set of superpixels, in which $sp_k = \{x_1^{(k)}, x_2^{(k)}, \ldots, x_{n_k}^{(k)}\}$ $(k = 1, 2, \ldots, K)$ is the kth superpixel, K is the number of superpixels, $x_i^{(k)}$ denotes the ith pixel in sp_k, and n_k is the number of pixels in sp_k.

Fisher score is one of the most widely used supervised FS methods [55]–[57] based on the Fisher theory (or criterion) [58], [59]. In general, fisher score uses the ratio of between-class scatter (which evaluates the separability of different classes) and within-class scatter (which models the compactness within each class) to measure the discrimination of the given feature according to the available labeled samples. In this article, an unsupervised trace ratio criterion is defined in the context of fisher score with superpixels. For HSI, all pixels within a superpixel are considered to belong to the same class. In order to model the compactness within each superpixel and evaluate the separability of different superpixels, the within-superpixel scatter matrix S_{WSP} and the between-superpixel scatter matrix S_{BSP} can be written with SP as follows:

$$S_{\mathrm{WSP}} = \sum_{k=1}^{K} \sum_{i=1}^{n_k} \left(x_i^k - m_k\right)\left(x_i^k - m_k\right)^T \tag{1}$$

$$S_{\mathrm{BSP}} = \sum_{k=1}^{K} n_k (m_k - m)(m_k - m)^T \tag{2}$$

where m is the mean vector of all the SP; m_k is the mean vector of the kth SP. The typical Fisher score is designed in the form of an unsupervised trace ratio criterion with SP. It employs the following optimization function as fitness function f:

$$f = \frac{\mathrm{trace}(S_{\mathrm{BSP}})}{\mathrm{trace}(S_{\mathrm{WSP}})} \tag{3}$$

where trace() denotes the trace of a matrix. S_{WSP} and S_{BSP} indicate the variability within each superpixel and between any two superpixels, respectively. However, the separability of different superpixels cannot be fully reflected only considering S_{WSP}. In order to take this issue into account, another fitness function is designed for considering the within-superpixel variability and the total variability simultaneously. To achieve this goal, the total-superpixel scatter matrix is defined as follows:

$$S_{\mathrm{TSP}} = \sum_{k=1}^{K} \sum_{i=1}^{n_k} \left(x_i^k - m\right)\left(x_i^k - m\right)^T \tag{4}$$

where $S_{\mathrm{TSP}} = S_{\mathrm{BSP}} + S_{\mathrm{WSP}}$.

The fitness function f can be equivalently rewritten as follows:

$$f = \frac{(S_{\mathrm{BSP}})}{\mathrm{trace}(S_{\mathrm{TSP}})} \tag{5}$$

in which the fitness function is optimized with small within-superpixel covariance and large total-superpixel covariance. A larger fitness value means a higher discriminative capability of the corresponding band subset.

C. Genetic Operations

In biological inheritance, genetic information passes from the current parent population to the members of the next generation through three main genetic operations, i.e., selection, crossover, and mutation. The searching process of the optimal solution in GA is achieved by using the above three genetic operations that imitate the evolutionary process of biology. Genetic operations guarantee the quality of the subset of bands selected by the GA-based method. In this article, we develop new genetic operations to guarantee during the BS process the selection of the subset of bands represented by chromosomes containing a small fixed number of selected bands having low redundancy.

1) Selection Operation: The selection operation selects some excellent individuals (i.e., chromosome) from the tth generation population $P(t)$ to the next generation $P(t+1)$ according to certain rules. The selection principle is based on the fitness value of individuals to avoid loss of the optimal solution and improve convergence speed. Various kinds of selection operations (e.g., proportional sampling random sampling, deterministic sampling, and Elitist model) have been applied in GAs. A proportional sampling-based selection operation, i.e., roulette wheel, is adopted in this article due to its simplicity. It uses the proportion of the fitness value of each individual to determine the selection possibility of every individual. The bigger the fitness value of an individual, the greater possibility of being selected, and vice versa. The selection probability of an individual can be computed as follows:

$$p_i = f_i \Big/ \sum_{j=1}^{NP} f_j \quad (i = 1, 2, \ldots, NP) \tag{6}$$

where f_i is the fitness value of the ith individual, p_i is the selection probability of the ith individual, and NP is the size of the population.

2) Crossover Operation: The crossover operation is the main process to produce new individuals in GAs. The selected individuals from the tth generation population $P(t)$ are randomly paired first. Then, for each pair of individuals, part of their chromosomes is exchanged with a certain probability, i.e., the crossover probability P_c. The searchability of GA can be improved rapidly by crossover operation.

In the proposed SSGA, the potential subset of bands represented by individuals should contain a small fixed number of selected bands. In this condition, common crossover operations, e.g., single-point crossover, multiple-point crossover, uniform crossover, and shuffle crossover are no longer applicable. Thus, a new crossover operation is developed. It maintains a fixed number of selected bands in the BS process and guarantees the desired number of bands.

Let $P(t) = \{c_1^{(t)}, c_2^{(t)}, \ldots, c_{NP}^{(t)}\}$ be the tth generation population by selection operation; $c_i^{(t)} = [0, 1, 0, \ldots, 1, \ldots, 1, 0]$

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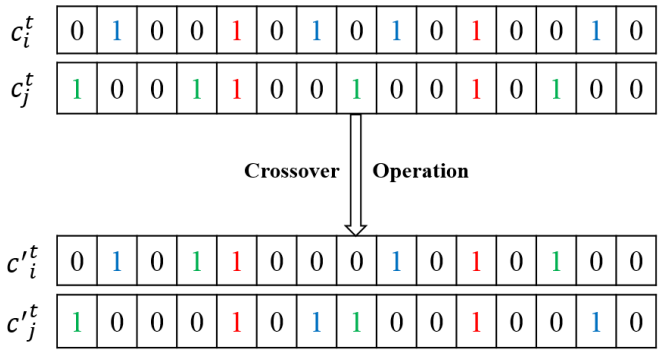


Fig. 2. Illustration of the proposed crossover operation.

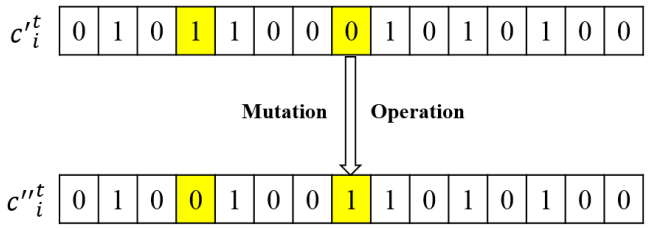


Fig. 3. Illustration of the proposed mutation operation.

indicates the ith chromosome (band subset) in the tth generation population. The length of $c_i^{(t)}$ is set to L. The number of genes valued 1 in $c_i^{(t)}$ is set to l, i.e., the desired number of selected bands. Fig. 2 illustrates the schematic of the proposed crossover operation. In the example of Fig. 2, L and l are assumed 15 and 6, respectively. Suppose $c_i^{(t)}$ and $c_j^{(t)}$ $(i \neq j)$ in a pair of randomly assigned chromosomes. If the genes with the value of 1 appear in the same location in $c_i^{(t)}$ and $c_j^{(t)}$, they are called "good features," whereas the other genes with the value of 1 are called "general features." In Fig. 2, two genes (red color) are "good features" and four genes (blue and green color) are "general features." In the proposed crossover operation, the "good features" are directly retained in the new individual and the "general features" in $c_i^{(t)}$ and $c_j^{(t)}$ are randomly crossed at 50%. New chromosomes $c_i'^{(t)}$ and $c_j'^{(t)}$ are obtained by the proposed crossover operation, and a new population $P'(t) = \{c_1'^{(t)}, c_2'^{(t)}, \ldots, c_{NP}'^{(t)}\}$ is generated for mutation operation. In this way, the desired number of selected bands can remain unchanged during the crossover process when retaining the "good features" and exchanging the "general features."

3) Mutation Operation: Mutation operation uses a small probability, i.e., mutation probability P_m, to change a gene value in one or more locations of a chromosome, generating a new individual. It can effectively increase the diversity of chromosomes in the population. In general mutation operation, one or more genes value on a chromosome may change. Taking single-point mutation as an example, if the value of a gene changes from 1 to 0 or 0 to 1, the number of selected bands will decrease or increase by 1 accordingly. It cannot satisfy the requirement of a desired and fixed number of selected bands.

To solve the above problem, a simple but effective mutation operation is presented. The proposed mutation operation can be regarded as the combination of two identical classical mutation operations. When chromosome mutates, one of the mutation operations is applied to genes with the value of 1, and the other one is applied to genes with the value of 0. As a result, the number of selected bands remains unchanged during the mutation process. Fig. 3 shows the procedure of the proposed mutation operation, in which the genes in $c_i'^{(t)}$ highlighted by yellow indicate genes to be mutated, $c_i''^{(t)}$ is the new chromosome from the mutation of $c_i'^{(t)}$, and $c_i''^{(t)}$ is one chromosome from $P''(t)$.

D. SSGA-Based BS

In the proposed SSGA approach, a spectral-spatial fitness function is designed by a superpixel-based unsupervised trace ratio criterion to measure the performance of the candidate subset of bands during the BS process. Moreover, the newly designed genetic operations are used to select a small fixed number of bands with low redundancy.

The candidate subset of bands in SSGA is constantly compared by continuous iterative optimization. A highly discriminative low redundancy band subset is selected with the termination criterion. Four common termination criteria are usually used in GA are as follows.

1) The number of generation reaches a predetermined value.
2) The fitness value remains constant during a specified number of generations (which can be viewed as the convergence of the GA).
3) The difference between the optimal fitness value of the current generation and the target fitness value is less than ε(ε is a small number).
4) The termination of GA is controlled by the user.

In this article, we combine 1) and 2) as the termination criterion of the proposed SSGA. *NG* is set to the maximum number of generations, and the proposed SSGA will be terminated when the fitness value remains the same within *NG*/2 generations. This can reduce the time complexity of the proposed SSGA.

The procedure of the proposed SSGA is illustrated in detail as follows.

III. Experimental Results and Discussion

A. Description of HSI Data Sets

In order to assess the effectiveness of the proposed SSGA method, three real-world hyperspectral data sets are used in the experiments. In the following, the description of these data sets is given.

Indian Pines [60]: The first data set was collected by the National Aeronautics and Space Administration using the Airborne Visible/Infrared Imaging Spectrometer sensor (AVIRIS). It is a subset of the Indian Pines North–South image gathered over northwest Indiana on June 12, 1992, and has 145 × 145 pixels with a spatial resolution at 20 m per pixel. The

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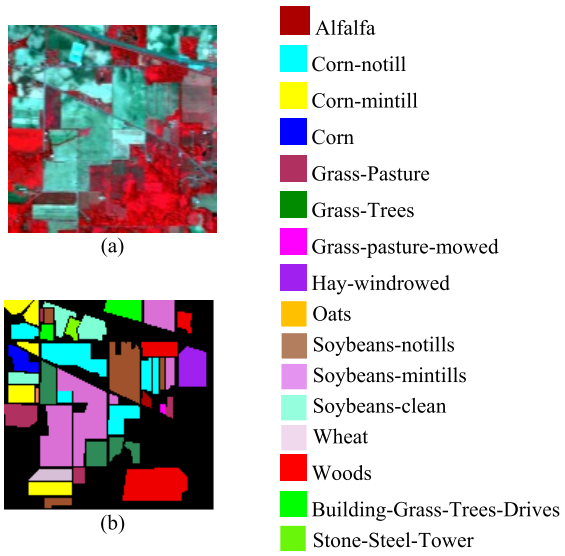


Fig. 4. (a) False-color composite of image of bands 57, 27, and 17. (b) Available ground truth map (Indian Pines data set).

original image contains 224 spectral channels in a wavelength range from 400 to 2500 nm. In general, 200 spectral bands are used in the experiments after removing four bad bands and 20 bands covering the region of water absorption (numbered 104–108, 150–163, and 220). The data set includes 16 land-cover classes, which are different types of crops and vegetation. The false-color composition of the image and the corresponding ground truth map are shown in Fig. 4.

Pavia University: The second data set was captured by the Reflective Optics System Imaging Spectrometer System (ROSIS) sensor surrounding the urban area of the University of Pavia during a flight campaign over Pavia, Northern Italy. This data set originally includes 115 bands with a spectral range of 430–860 nm. Twelve noise bands were discarded, resulting in 103 spectral bands remain for the experiment. The image has a size of 610 × 340 pixels and a high spatial resolution (1.3 m per pixel) with a well-defined spatial texture. Nine land-cover classes, which are trees, asphalt, bitumen, gravel, metal sheets, shadow, bricks, meadows, and bare soil were considered. Fig. 5 displays the false-color composition image and the corresponding ground truth map of this data set.

Pavia Center: The third data set was obtained by the ROSIS sensor with 115 spectral channels over the center of Pavia, Italy. After removing the low SNR bands, 102 bands were used in the experiments. The image scene has a size of 1096 × 1096 pixels with a spatial resolution of 1.3 m. Nine classes of interest are present in this image. Fig. 6 shows the false-color image of this image and the nine ground-truth classes of interest.

B. Experimental Setup

1) Compared Methods: The experiments are conducted by comparing the proposed SSGA with traditional GA-based BS method and seven state-of-the-art unsupervised BS methods, including two ranking-based techniques (the ID [20] and the MV-PCA [21]) and four clustering-based techniques (the AP [24], the AAP [26], the SML-AP, and the SRL-AP [27]) and a ranking-based clustering methods (the E-FDPC [28]). These BS algorithms compared in the experiments are briefly described as below.

Algorithm Proposed SSGA Band Selection

Input:

Hyperspectral images data set, $X = \{x_1, x_2, \ldots, x_L\} \subset \mathbb{R}^{N\times L}$

Number of spectral bands, L

Desired number of selected bands, l

Number of superpixels, K

Max generation, GN

Size of population, NP

Crossover probability, P_c

Mutation probability, P_m

Output:

Representative band subset $Y = \{y_1, y_2, \ldots, y_l\}$, $y_l \in X$, $(l \ll L)$

Procedure:

1) Initialize code chromosome c_i
2) with the desired number of spectral bands l, i.e., a binary vector contains l zero denoting a candidate solution.
3) Extend ERS-based superpixel to HSI pixel vector, i.e., obtain the $sp_k = \{x_1^{(k)}, x_2^{(k)}, \ldots, x_{n_k}^{(k)}\}$ $(k = 1, 2, \ldots, K)$.
4) Establish the population $P = \{c_1, c_2, \cdots, c_i, \cdots, c_{NP}\}$ with NP chromosomes (i.e., individuals), and set the generation number $GN = 0$.
5) Calculate the fitness value of each candidate solution using (5).
6) Perform the genetic operations, i.e., selection operation, crossover operation, and mutation operation.
7) Calculate the fitness value of each candidate solution obtained in step 5. Update $GN = GN + 1$. If the termination criterion is satisfied, go to step 7; otherwise, go to step 5.
8) Choose the optimal candidate solution based on fitness values. The final selected band subset Y contained in the optimal solution and the number of bands l can be used for HSI classification.

1) *GA:* The band subset is selected with the typical genetic operation and the same fitness function as the SSGA.
2) *ID [20]:* The NG is used to measure the prioritization of bands.
3) *MV-PCA [21]:* The bands are prioritized with the maximum variance criterion.
4) *AP [24]:* Negative Euclidean distance and median similarity are adopted as the similarity measure between two different bands and self-similarity (i.e., preference), respectively.

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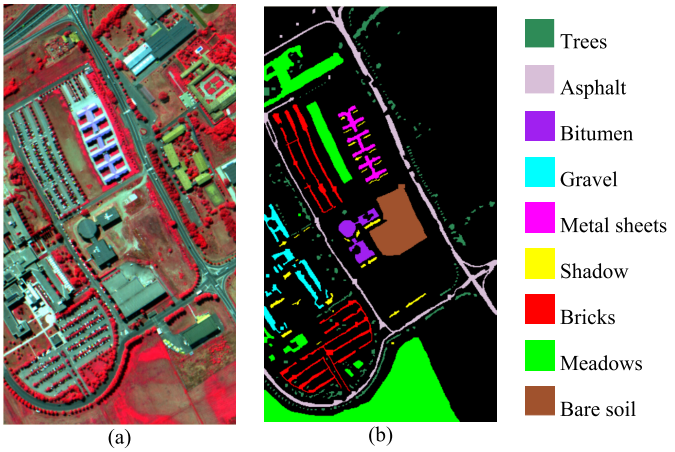


Fig. 5. (a) False-color composite image of bands 102, 56, and 31. (b) Available ground truth map (University of Pavia data set).

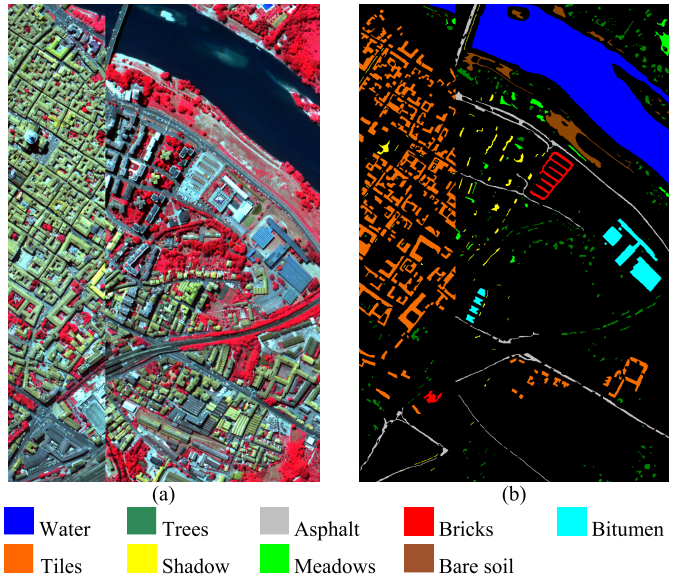


Fig. 6. (a) False-color composite image of bands 102, 56, and 31. (b) Available ground truth map (Pavia Center data set).

5) *AAP [26]:* AP with negative SAM, and a fixed number of selected bands obtained by an exemplar number determination procedure.
6) *SML-AP and SRL-AP [27]:* Two unsupervised BS methods derived from the same learning framework. The SML-AP and SRL-AP measure bands based on metric learning and representation learning, respectively, then select bands with the AP.
7) *E-FDPC [28]:* The score of each band is calculated by weighting the normalized local density and the intracluster distance obtained by ranking-based clustering method, i.e., a FDPC algorithm.

In addition, the results obtained by using all of the original spectral bands (Baseline) are also computed for comparison in the experiments.

2) Quality Indices:

a) Accuracy measures: To assess the performance of the proposed SSGA approach, a classical classification algorithm [i.e., support vector machine (SVM)] is applied to the selected subset of bands of all the considered BS methods. In our experiments, the SVM is implemented with the LibSVM[1] toolbox with the radial basis function (RBF), and the parameters *C* and *gamma* were optimized by fivefold cross-validation.

For Indian Pines, Pavia University, and Pavia Center data sets, 5%, 2%, and 1% of labeled pixels were randomly selected for the definition of the training sets, and the remaining samples were used as test samples. The detailed information of training and test samples with land-cover classes on the three considered data sets is given in Table I. To reduce the influence

TABLE I
NUMBER OF TRAINING AND TEST SAMPLES IN EACH LAND-COVER CLASS FOR INDIAN PINE, PAVIA UNIVERSITY, AND PAVIA CENTER DATA SETS

Data sets	Class	Land Cover Type	Training Samples	Test Samples
Indian Pines	1	Alfalfa	3	51
	2	Buildings-Grass-Trees-Drives	19	361
	3	Corn	12	222
	4	Corn-min Till	42	792
	5	Corn-no Till	72	1362
	6	Grass-pasture-mowed	2	24
	7	Grass/Pasture	25	472
	8	Grass/Trees	38	709
	9	Hay-windrowed	25	464
	10	Oats	1	19
	11	Soybean-clean Till	31	583
	12	Soybean-min Till	124	2344
	13	Soybean-no Till	49	919
	14	Stone-Steel-Towers	5	90
	15	Wheat	11	201
	16	Woods	65	1229
	Total		524	9842
Pavia University	1	Asphalt	133	6498
	2	Meadows	373	18276
	3	Gravel	42	2057
	4	Trees	62	3002
	5	Metal sheets	27	1318
	6	Bare soil	101	4928
	7	Bitumen	27	1303
	8	Bricks	74	3608
	9	Shadows	19	928
	Total		858	41918
Pavia Center	1	Water	660	65311
	2	Trees	76	7522
	3	Asphalt	31	3059
	4	Self-blocking bricks	27	2658
	5	Bitumen	66	6518
	6	Tiles	93	9155
	7	Shadow	73	7214
	8	Meadows	429	42397
	9	Bare soil	29	2834
	Total		1484	146668

[1]http://www.csie.ntu.edu.tw/~cjlin/libsvm/

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TABLE II
BASIC PARAMETERS OF THE PROPOSED SSGA

Nomenclature	Interpretation	Value
NG	Max generation	100
NP	Size of population	100
P_c	Crossover probability	0.9
P_{m1}	Initial mutation probability	0.02
P_{m2}	Ultimate mutation probability	0.1

of random effects, the training and test data sets are randomly chosen from the data five times. The classification performance is assessed by the overall accuracy (OA), the average OA (AOA), and the kappa coefficient (κ). Moreover, the standard deviation (SD) of AOA over the five experiments is also reported.

b) Number of selected bands: For the three HSI data sets, the desired number of selected bands is usually not known *a priori*. For demonstrating the effectiveness of proposed SSGA, experiments with the different numbers of selected bands (in the range between 5 and 60) were carried out for all methods on each data set, except the GA method in which the optimal number of selected bands is automatically determined.

3) Parameters Setting of the SSGA: The basic parameters of the SSGA used in the experiments are listed in Table II. It should be noted that the mutation probability varying with generation in a linear relation is adopted. Thus, the actual mutation probability P_m of ith generation can be computed as follows:

$$P_m = P_{m1} + i/NG \times (P_{m2} - P_{m1}) \quad (i = 1, 2, \ldots, NG). \quad (7)$$

For the SSGA, $NB = l$ can be set as the desired number of selected bands.

C. Results

1) Effect of the Number of Superpixel on the SSGA: For the proposed SSGA, the desired properties of a superpixel are that only one ground object should be contained in it and most of the spatial structure information should be preserved. The number of superpixels can be selected based on ground object distribution and structural texture complexity in the three considered HSI data sets. In the experiments, the range of the number of superpixels used is {300, 500, 700, 900} for the Indian Pines, University of Pavia, and Pavia Center data sets.

The influence of the number of superpixels on the SSGA is analyzed from two perspectives, i.e., classification performance and distribution of selected bands. The classification accuracy obtained by the proposed SSGA with different numbers of superpixels is shown in Fig. 7. From Fig. 7, one can see that the AOA curves have only a slight fluctuation versus the number of superpixels for the three considered HSI data sets. Fig. 8 shows the effect of the number of superpixels considered in the SSGA on the distribution of the selected subset of bands. From Fig. 8, one can see that the distribution of bands selected has a high consistency among adjacent spectral channels in the three HSI data sets with different numbers of superpixels. This indicates that the proposed SSGA is robust to the number of superpixels considered. This is probably due to the fact that the superpixel is used to define the fitness function in an unsupervised way. The proposed SSGA aims to find the optimal subset of bands that enables the fitness value to be optimized based on the homogeneity in superpixel. As the superpixel algorithm is inherently an over-segmentation method, the homogeneity of superpixels can be guaranteed starting from a given number of superpixels.

Due to the difference of spatial resolution and the distribution of ground objects present in the three HSI data sets, the AOA values of the SSGA have different behaviors compared with the Baseline versus the number of superpixels. The Indian Pines data set has a relatively small size and low spatial resolution; thus, the land-cover regions are difficult to distinguish. Accordingly, a small number of superpixels (i.e., 300) are sufficient for maintaining the homogeneity within the same ground objects. The University of Pavia data have a high spatial resolution and many small size ground objects. Thus, a large number of superpixels (i.e., 700) is set for this data set. The Pavia Center data set also has high spatial resolution with relatively concentrated structures and textures. Accordingly, the number of superpixels is set to 500 in the following experiments.

2) SSGA Versus GA: In this section, a comparative experiment between the proposed SSGA and the GA is given. It is worth noting that we set the same basic parameters (see Table II), the fitness function, and the number of superpixels for the GA and the proposed SSGA on the three considered data sets. For the SSGA and the GA, five BS trials were carried out owing to the multiple optimal solutions, i.e., the different numbers of selected bands would be obtained in the BS process. Table III lists the results obtained by the proposed SSGA and the GA. Two BS cases of the SSGA, i.e., the minimal number of selected bands with classification performance not lower than GA and the same number of selected bands with GA, are considered. From Table III, one can observe that in most cases, the proposed SSGA achieves higher classification accuracies than the GA in the two considered BS cases on the three HSI data sets. In the first case, the number of bands selected by the GA is much higher (about 3–5 times higher) than those of the proposed SSGA. Meanwhile, the mean AOAs obtained by the proposed SSGA are always not lower than those of the GA. For the second case, one can see that the mean AOAs achieved by the SSGA are 2.56%, 7.05%, and 0.29% higher than those of the GA with the same number of selected bands. The execution times of the GA and the proposed SSGA under the two BS cases are also reported in Table III. In the first case, as the number of bands selected by the SSGA is much smaller than those selected by the GA, the execution times required by the SSGA on the three HSI data sets are far less than those taken by the GA. In the second case, for Indian Pines and Pavia University data sets, one can see that the execution times of the SSGA are slightly higher than those of the GA. This is because the constraint manipulations adopted by the modified

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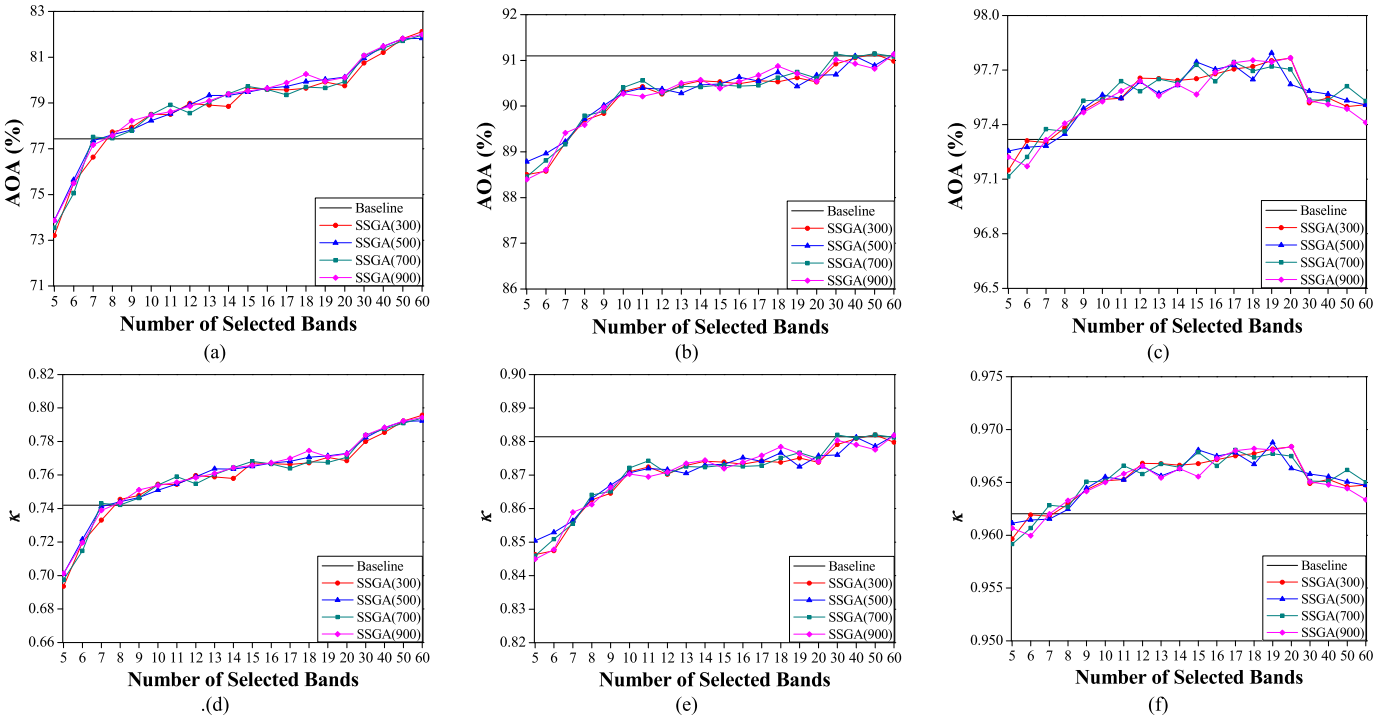


Fig. 7. AOA and κ versus the number of selected bands obtained by the proposed SSGA with different numbers of superpixel. The AOA and κ results achieved by using all spectral channels are also reported (Baseline). (a) AOA on Indian Pines data set. (b) AOA on Pavia University data set. (c) AOA on Pavia Center data set. (d) κ on Indian Pines data set. (e) κ on Pavia University data set. (f) κ on Pavia Center data set.

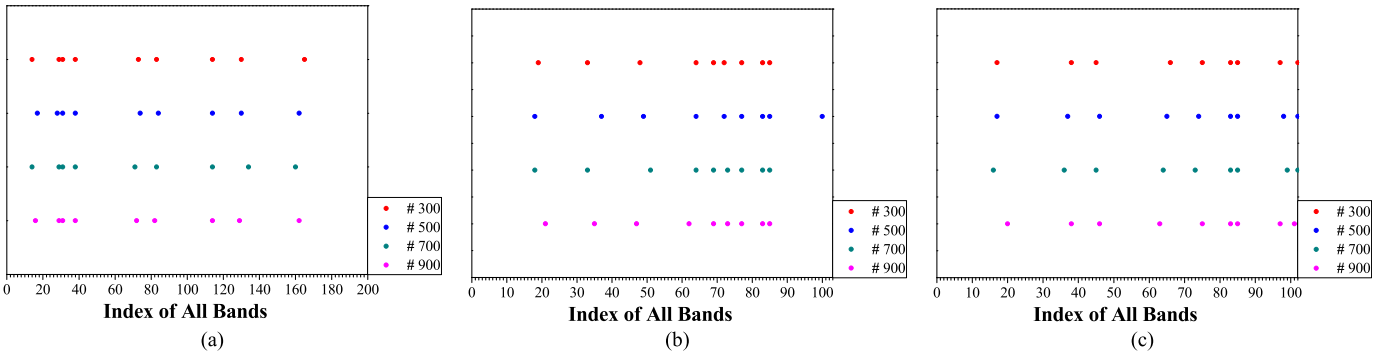


Fig. 8. Distribution of the selected subset of bands obtained by the proposed SSGA with different numbers of superpixels on (a) Indian Pines, (b) Pavia University, and (c) Pavia Center data sets.

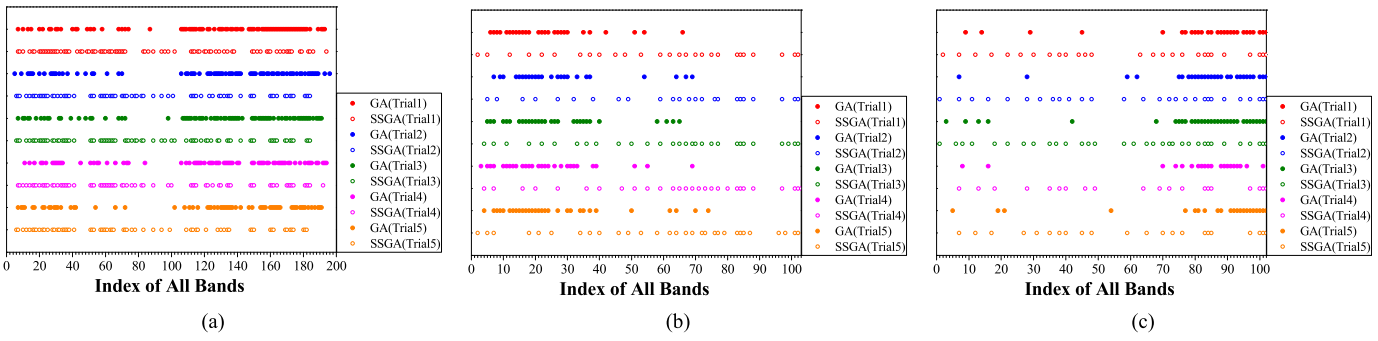


Fig. 9. Distribution of the selected subset of bands obtained by the proposed SSGA and the GA in five trials on (a) Indian Pines, (b) Pavia University, and (c) Pavia Center data sets.

genetic operations in the SSGA require an additional time compared with the GA. Nevertheless, for the Pavia Center data set, the execution times of the SSGA are smaller than those of the GA. This is due to the faster convergence speed of the proposed SSGA compared with the GA on this data set. Considering both classification accuracy and execution

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TABLE III

RESULTS OBTAINED BY THE PROPOSED SSGA AND THE STANDARD GA ON THE INDIAN PINE, THE PAVIA UNIVERSITY, AND THE PAVIA CENTER DATA SETS

Data sets	Methods		Trial1	Trial2	Trial3	Trial4	Trial5	Mean	SD
Indian Pines	GA	Number of selected bands	88	78	81	75	67	—	—
		AOA(%)	79.06	79.31	79.51	79.67	79.42	79.40	0.19
		κ	0.7603	0.7636	0.7661	0.7676	0.7647	0.7645	0.0023
		Time(second)	531.43	470.86	522.32	478.51	464.72	493.57	27.69
	SSGA	Number of selected bands	15					—	—
		AOA(%)	79.45	79.82	79.61	79.74	79.66	**79.65**	0.11
		κ	0.7649	0.7691	0.7667	0.7681	0.7676	**0.7673**	0.0013
		Time(second)	49.80	50.34	57.23	56.87	51.61	**53.17**	3.22
		Number of selected bands	88	78	81	75	67	—	—
		AOA(%)	82.03	81.80	81.58	82.08	82.32	**81.96**	0.23
		κ	0.7948	0.7925	0.7896	0.7954	0.7983	**0.7941**	0.0027
		Time(second)	607.21	581.35	590.26	568.43	527.62	574.97	26.83
Pavia University	GA	Number of selected bands	27	24	29	28	30	—	—
		AOA(%)	81.80	85.59	81.99	84.85	86.35	84.12	1.71
		κ	0.7494	0.8055	0.7515	0.7961	0.8161	0.7837	0.0255
		Time(second)	851.59	843.65	871.78	853.46	867.22	857.54	10.41
	SSGA	Number of selected bands	5					—	—
		AOA(%)	88.63	88.16	88.62	88.51	88.34	**88.45**	0.16
		κ	0.8485	0.8418	0.8483	0.8467	0.8444	**0.8459**	0.0023
		Time(second)	271.03	269.32	270.84	270.79	270.33	**270.46**	0.62
		Number of selected bands	27	24	29	28	30	—	—
		AOA(%)	91.14	91.33	91.34	91.17	90.86	**91.17**	0.16
		κ	0.8820	0.8846	0.8848	0.8823	0.8784	**0.8824**	0.0021
		Time(second)	946.23	919.48	1039.12	960.26	921.53	957.32	43.67
Pavia Center	GA	Number of selected bands	27	28	33	20	22	—	—
		AOA(%)	97.37	97.44	97.44	96.96	97.20	97.28	0.17
		κ	0.9628	0.9638	0.9638	0.9569	0.9603	0.9615	0.0024
		Time(second)	2751.55	2755.28	2813.06	2729.19	2838.13	2777.44	41.09
	SSGA	Number of selected bands	7					—	—
		AOA(%)	97.33	97.25	97.31	97.24	97.29	**97.28**	0.03
		κ	0.9621	0.9610	0.9619	0.9610	0.9616	**0.9615**	0.0004
		Time(second)	864.80	910.76	659.18	935.62	940.45	**862.16**	104.97
		Number of selected bands	27	28	33	20	22	—	—
		AOA(%)	97.53	97.50	97.50	97.60	97.72	**97.57**	0.08
		κ	0.9650	0.9646	0.9646	0.9660	0.9678	**0.9656**	0.0011
		Time(second)	2433.58	2571.96	2240.53	1662.86	1997.76	**2181.34**	323.18

time, we can conclude that the proposed SSGA shows better performance than the GA.

Fig. 9 shows the distribution of band subsets selected by the proposed SSGA and the GA on the three HSI data sets. One can observe from Fig. 9(a) that the bands selected by the GA on Indian Pines data set are concentrated in the range of 110–190 on five trials. On the contrary, the bands selected by the SSGA are more evenly distributed across the range of all bands. For the Pavia University and Pavia Center data sets [Fig. 9(b) and (c)], it is clear that the bands selected by the GA are mainly distributed in the top 40 bands and the last 30 bands, respectively, whereas the bands selected by SSGA are more dispersed, covering the whole spectrum. One can see from the result of this experiment that the proposed SSGA approach selects band subsets less redundant and more informative. The comparison of results in terms of AOAs and the number of selected bands obtained on the three HSI data sets shows the improved effectiveness of the proposed SSGA approach.

3) Comparison With State-of-the-Art Methods: In this experiment, the classification performances of the proposed SSGA are compared with seven state-of-the-art methods (i.e.,

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ID, MV-PCA, E-FDPC, AP, AAP, SML-AP, and SRL-AP) and the Baseline. Fig. 10 shows the AOA and κ results of all methods with the SVM on the three HSI data sets. The best results of the proposed SSGA with 300, 700, and 500 superpixels for the Indian Pines, Pavia University, and Pavia Center data sets, respectively, are shown in Fig. 10. From Fig. 10, one can observe that the proposed SSGA approach achieved the highest classification accuracies compared with other methods when using the same number of selected bands.

For the experimental results on the Indian Pines data set, from Fig. 10(a) and (d), one can observe that the proposed SSGA exhibits the best performance compared with the seven state-of-the-art methods. The AOA and κ obtained by the proposed SSGA maintain steady growth with the increase of the number of selected bands. It is worth noting that the SSGA achieves a higher AOA and κ (78.18% and 0.7507) with only eight selected bands than the Baseline (77.43% and 0.7420) with all 200 bands. It is expected due to the curse of dimensionality problem. Among the compared methods, only SML-AP can obtain a better result than the Baseline with 13 selected bands, but the accuracies obtained by SML-AP decrease with the number of selected bands beyond 30. The MV-PCA yields the worst classification results. In terms of SD [Fig. 11(a) and (d)], the SSGA produced more stable results than the other compared methods.

For the experimental results of the Pavia University data set, from Fig. 10(b) and (e), the proposed SSGA achieves the best classification performance and the optimal SD. The accuracies obtained by the SSGA are much higher than those obtained by other considered methods when the number of selected bands is small. The AOA and κ obtained by the SSGA are higher than those by Baseline when the number of selected bands reaches 30 and remain stable. The SML-AP achieves only one time slightly higher AOA and κ (0.07% and 0.0010) with 50 selected bands than the proposed SSGA. The MV-PCA and E-FDPC show a similar trend of AOA and κ when increasing the number of selected bands. The SML-AP and the SRL-AP express instability in both AOA and κ curves. The ID obtained the lowest classification accuracies among the eight considered BS algorithms.

For the experimental results of the Pavia Center data set, from Fig. 10(c) and (f), the proposed SSGA also achieves the highest AOA and κ and maintains relatively high stability [Fig. 11(c) and (f)]. In greater detail, the SSGA generates AOA value of 97.33% and κ value of 0.9622 with five selected bands, which are higher than those achieved by Baseline (97.31% and 0.9620). The E-FDPC, SML-AP, and SRL-AP achieve several times higher accuracies than the proposed SSGA, but only with more than 20 selected bands. The ID and the MV-PCA show poor performance when the number of selected bands is small. They can get a better performance than the Baseline only when the number of selected bands is larger than 16 and 40, respectively. Compared with the first two data sets, the AOA and κ values of the proposed SSGA are significantly higher and the performance is slightly superior to those of the other methods when increasing the number of selected bands. This may be due to the kind of morphology of the landscape in the Pavia Center data set, i.e., the regular and massive homogeneous land-cover makes this data set more easily to be classified.

TABLE IV

SD OF AOA AND κ OBTAINED BY THE PROPOSED SSGA IN DIFFERENT BS TRIALS ON THE INDIAN PINE, THE PAVIA UNIVERSITY, AND THE PAVIA CENTER DATA SETS

Datasets		Indian Pines		Pavia University		Pavia Center	
		AOA(%)	κ	AOA(%)	κ	AOA(%)	κ
Number of Selected Bands	5	0.52	0.0061	0.18	0.0025	0.05	0.0007
	6	0.63	0.0071	0.23	0.0031	0.05	0.0007
	7	0.54	0.0062	0.36	0.0050	0.03	0.0005
	8	0.22	0.0026	0.30	0.0040	0.04	0.0006
	9	0.17	0.0020	0.24	0.0032	0.08	0.0011
	10	0.32	0.0038	0.14	0.0019	0.08	0.0012
	11	0.39	0.0046	0.18	0.0024	0.07	0.0010
	12	0.29	0.0034	0.31	0.0042	0.13	0.0018
	13	0.28	0.0033	0.23	0.0031	0.06	0.0008
	14	0.51	0.0060	0.25	0.0034	0.07	0.0010
	15	0.13	0.0014	0.25	0.0033	0.08	0.0011
	16	0.34	0.0039	0.16	0.0021	0.07	0.0010
	17	0.23	0.0027	0.13	0.0017	0.05	0.0007
	18	0.34	0.0041	0.08	0.0011	0.11	0.0015
	19	0.17	0.0020	0.35	0.0047	0.03	0.0004
	20	0.13	0.0016	0.23	0.0031	0.11	0.0016
	30	0.21	0.0024	0.17	0.0022	0.08	0.0011
	40	0.31	0.0035	0.09	0.0013	0.12	0.0017
	50	0.15	0.0018	0.17	0.0023	0.05	0.0007
	60	0.18	0.0021	0.10	0.0013	0.14	0.0020
Mean		**0.30**	**0.0035**	**0.21**	**0.0028**	**0.07**	**0.0011**

D. Discussion

According to the experimental results presented in Section III-C3, one can see that the proposed SSGA almost always outperforms the other BS algorithms in terms of AOA and κ. In this section, the stability and effectiveness of the SSGA are analyzed.

1) Stability Analysis: In the BS process, the proposed SSGA can select the desired number of suboptimal bands using a robust way in a reduced search space, which is different from the traditional GA searching in the full bands combination space. Similar to the GA-based method, the subset of bands selected by the SSGA is not guaranteed to be exactly the same when the same number of bands is selected. In order to further analyze the stability of the proposed SSGA, the classification performance and the distribution of five independent BSs with the same selected bands are considered. The classification results with five independent BS trials on the three HSI data sets are shown in Fig. 12, and the SD of classification accuracies among different trials is listed in Table IV. From Fig. 12, one can observe that the classification accuracies of the different selected bands for all three data sets are close to each other. From Table IV, one can see that the mean SDs of AOA with five independent BSs are 0.3%, 0.21%, and 0.07%, and the mean SDs of κ are 0.0035, 0.0028, and 0.0011 on

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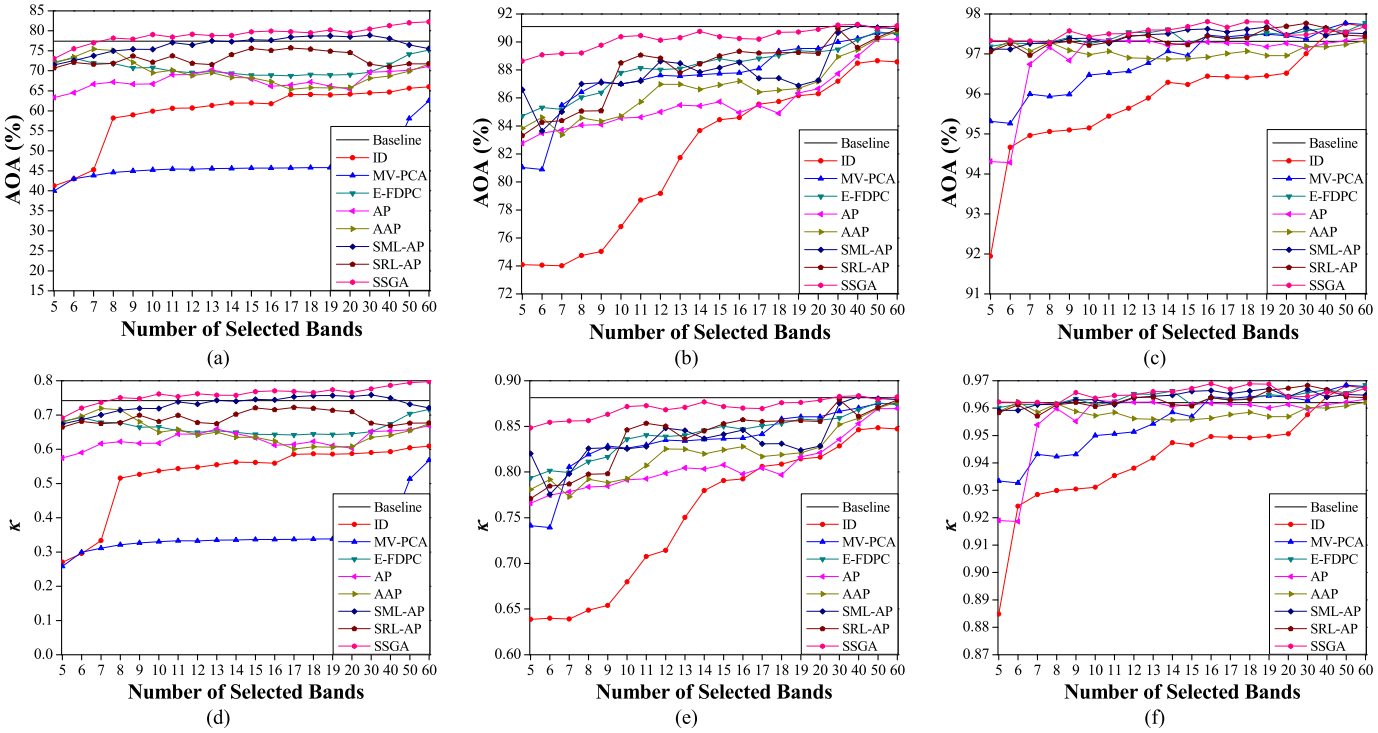


Fig. 10. AOA and κ versus the number of selected bands obtained by the ID, MV-PCA, E-FDPC, AP, AAP, SML-AP, SRL-AP, and the proposed SSGA methods. The AOA and κ results achieved by using all spectral channels are also reported (Baseline). (a) AOA on Indian Pines data set. (b) AOA on Pavia University data set. (c) AOA on Pavia Center data set. (d) κ on Indian Pines data set. (e) κ on Pavia University data set. (f) κ on Pavia Center data set.

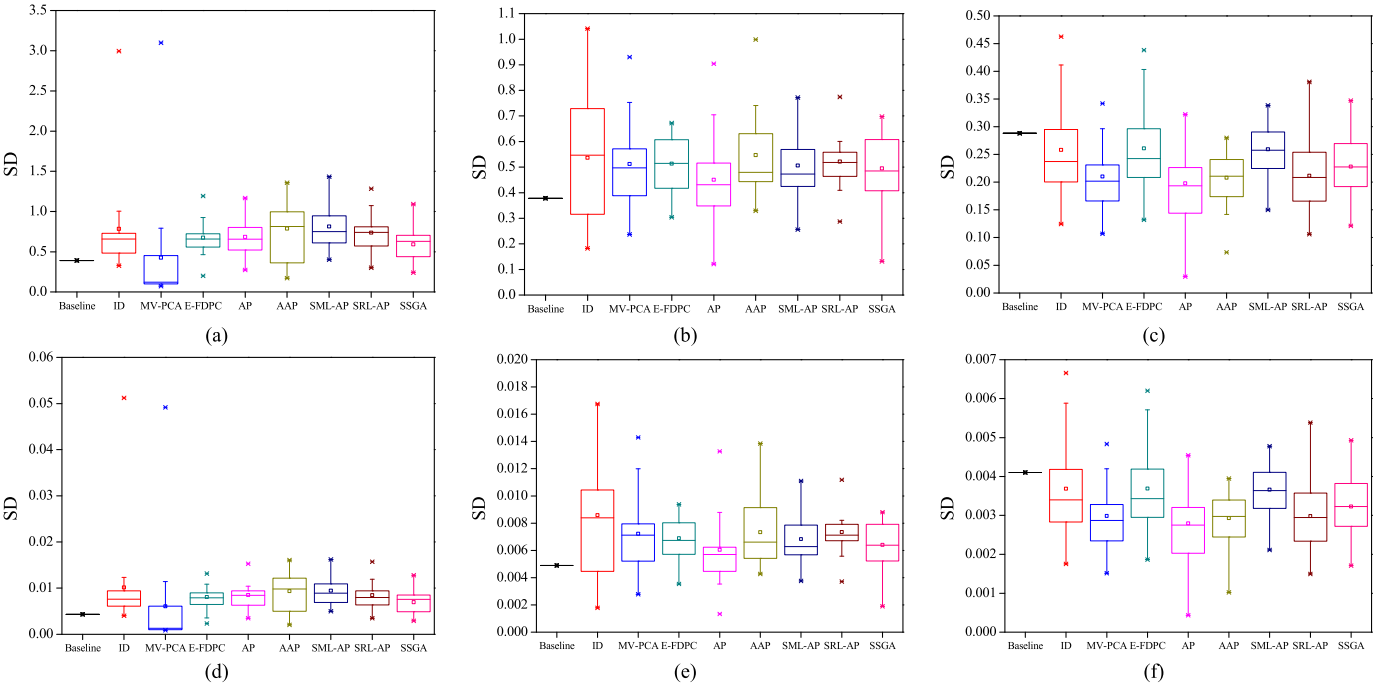


Fig. 11. SD of the ID, MV-PCA, E-FDPC, AP, AAP, SML-AP, SRL-AP, and the proposed SSGA methods. The SD results achieved by using all spectral channels are also reported (Baseline). (a) SD of AOA on Indian Pines data set. (b) SD of AOA on Pavia University data set. (c) SD of AOA on Pavia Center data set. (d) SD of κ on Indian Pines data set. (e) SD of κ on Pavia University data set. (f) SD of κ on Pavia Center data set.

the Indian Pines, Pavia University, and Pavia Center data sets, respectively. Table V lists the results of five independent BSs with nine selected bands. From Table V, for the Indian Pines data set, one can see that the bands selected in the first, the third, and the fifth trials are very close to each other, and the bands selected in the second and the fourth trials can be considered equally representative. This may be due to the fact that the ground objects have very similar

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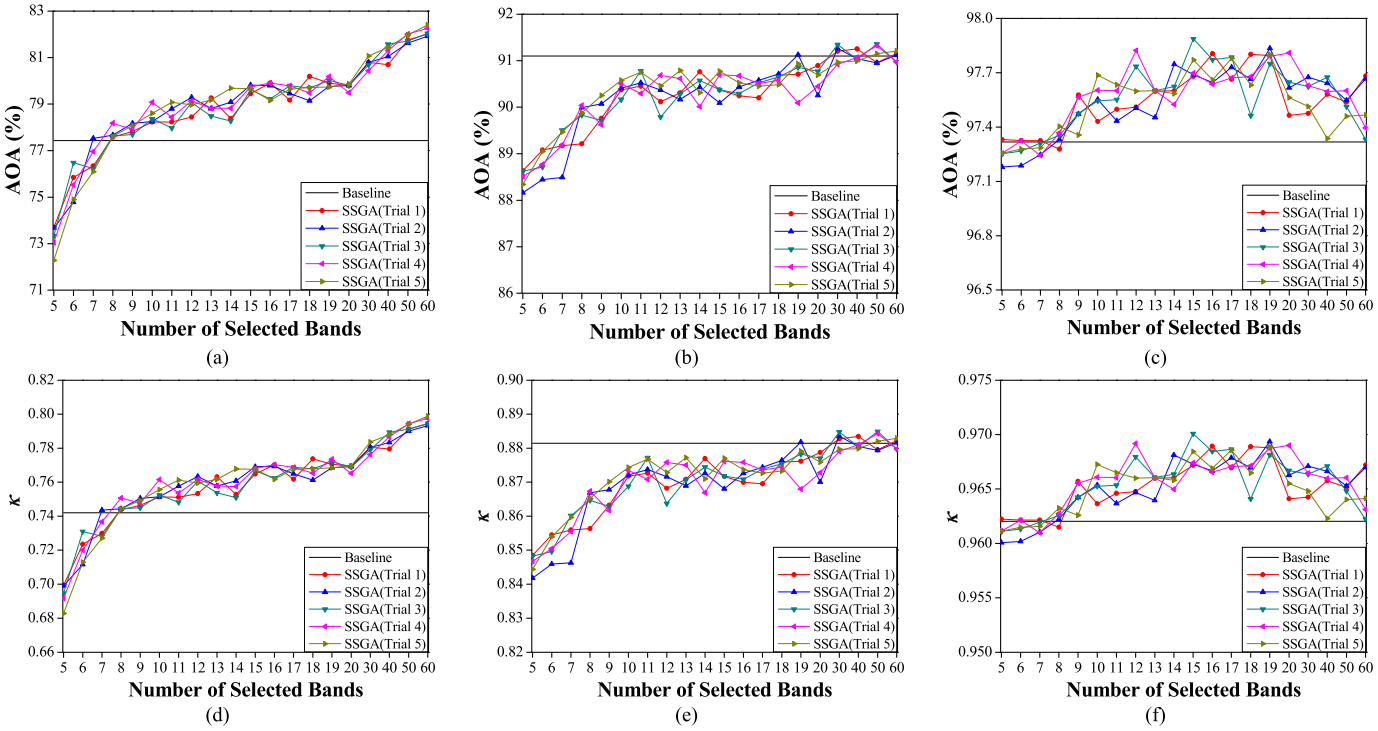


Fig. 12. AOA and κ versus the number of selected bands obtained by the proposed SSGA on five independent BS trials. The AOA and κ results achieved by using all spectral channels are also reported (Baseline). (a) AOA on Indian Pines data set. (b) AOA on Pavia University data set. (c) AOA on Pavia Center data set. (d) κ on Indian Pines data set. (e) κ on Pavia University data set. (f) κ on Pavia Center data set.

reflection characteristics among adjacent spectra. In greater detail, comparing bands selected in the first and the fourth trials, only one band is different (band 64 and band 83 located in the first and the fourth trials, respectively). For the Pavia University and Pavia Center data sets, a similar conclusion can be drawn from the selected bands listed in Table V. These results demonstrate that the proposed SSGA approach can perform a stable BS.

2) Effectiveness Analysis: A statistical significance analysis about the effectiveness of the proposed SSGA is performed in this section. Compared with other reference BS methods, the relatively high classification results of the SSGA with nine selected bands on the three HSI data sets are chosen. McNemar's test [61], a statistical test based on the standardized normal test statistic, is used to evaluate the statistical significance of the difference between selection methods based on the AOAs of different BS methods with the same classifier and labeled samples, which is defined as

$$Z = \frac{t_{12} - t_{21}}{\sqrt{t_{12} - t_{21}}} \tag{8}$$

where t_{ij} denotes the number of pixels correctly classified by learner i (i.e., BS method i) but misclassified by learner j (i.e., BS method j), and Z indicates the statistical significance of the difference between ith and jth BS methods. The difference between ith and jth BS methods is considered as statistically significant at the 95% level of confidence if $|Z| > 1.96$. The results of McNemar's test on the proposed SSGA and other reference algorithms for the three HSI data sets are listed in Table VI. According to the mean values on five BS trials of McNemar's test (i.e., the last column of Table VI) for the three data sets, one can see that the proposed SSGA is always superior ($Z > 1.96$) to other considered reference methods. The results of McNemar's test demonstrate that the

TABLE V
BANDS SELECTED BY THE SSGA WITH FIVE INDEPENDENT BS TRIALS ON THE INDIAN PINE, THE PAVIA UNIVERSITY, AND THE PAVIA CENTER DATA SETS

Datasets	# Trial	Selected Bands
Indian Pines	Trial1	14, 28, 31, 37, 64, 73, 114, 131, 163
	Trial2	18, 29, 31, 38, 74, 82, 115, 126, 164
	Trial3	20, 29, 35, 38, 62, 70, 113, 130, 162
	Trial4	14, 29, 31, 38, 73, 83, 114, 130, 165
	Trial5	18, 29, 35, 41, 63, 72, 114, 133, 162
Pavia University	Trial1	21, 35, 51, 65, 72, 77, 83, 85, 100
	Trial2	17, 35, 48, 62, 69, 72, 78, 83, 85
	Trial3	17, 35, 50, 64, 69, 72, 77, 83, 85
	Trial4	18, 33, 51, 64, 69, 73, 77, 83, 85
	Trial5	20, 33, 48, 62, 69, 76, 83, 85, 97
Pavia Center	Trial1	16, 36, 47, 66, 75, 83, 85, 100, 102
	Trial2	17, 37, 45, 67, 76, 83, 85, 97, 102
	Trial3	17, 37, 46, 66, 76, 83, 85, 99, 102
	Trial4	17, 37, 46, 65, 74, 83, 85, 98, 102
	Trial5	18, 37, 46, 67, 76, 83, 86, 99, 101

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TABLE VI

COMPUTING TIME (IN SECONDS) TAKEN BY THE PROPOSED SSGA AND THE OTHER CONSIDERED METHODS WITH 9 SELECTED BANDS ON THE INDIAN PINE, THE PAVIA UNIVERSITY, AND THE PAVIA CENTER DATA SETS

Datasets	Methods	SSGA					
		Z (Trial1)	*Z* (Trial2)	*Z* (Trial3)	*Z* (Trial4)	*Z* (Trial5)	*Z* (Mean)
Indian Pines	ID	32.25	33.74	29.00	32.14	32.94	32.01
	MV-PCA	50.43	49.53	46.65	47.61	49.76	48.79
	E-FDPC	15.51	17.04	12.72	12.80	15.63	14.74
	AP	23.77	26.66	19.85	20.10	24.87	23.05
	AAP	18.25	17.78	8.29	11.61	12.33	13.65
	SML-AP	7.53	3.45	0.29	7.09	8.42	5.36
	SRL-AP	10.22	10.45	4.79	8.47	14.53	9.69
Pavia University	ID	55.60	63.66	63.38	70.12	63.27	63.21
	MV-PCA	12.81	29.78	21.28	6.06	15.82	17.15
	E-FDPC	15.93	20.06	22.20	11.62	23.27	18.61
	AP	29.27	28.43	36.44	30.00	36.10	32.05
	AAP	27.32	27.30	33.88	27.76	24.51	28.15
	SML-AP	12.99	14.35	18.98	8.29	13.60	13.64
	SRL-AP	24.05	28.21	30.60	17.41	23.65	24.78
Pavia Center	ID	37.41	45.80	36.30	58.81	52.12	46.09
	MV-PCA	24.76	36.47	26.31	44.72	37.16	33.88
	E-FDPC	1.62	3.43	5.59	9.88	0.86	4.28
	AP	17.58	15.84	7.10	17.69	18.89	15.42
	AAP	9.15	7.50	0.06	11.84	18.14	9.34
	SML-AP	7.11	1.54	4.76	1.84	0.73	3.20
	SRL-AP	3.88	3.03	2.38	8.73	3.56	4.32

TABLE VII

MCNEMAR'S TEST BETWEEN THE PROPOSED SSGA AND THE ID, MVPCA, E-FDPC, AP, AAP, SML-AP, AND SRL-AP ON THE INDIAN PINE, THE PAVIA UNIVERSITY, AND THE PAVIA CENTER DATA SETS

Methods		ID	MV-PCA	E-FDPC	AP	AAP	SML-AP	SRL-AP	SSGA
Datasets	Indian Pines	1.31	0.20	0.44	1.28	1.09	4.46	4.62	47.56
	Pavia University	1.10	0.23	0.67	0.75	0.45	3.03	3.10	279.33
	Pavia Center	3.28	0.73	3.64	3.51	1.33	10.14	10.47	886.65

band subsets selected by the proposed SSGA approach are more discriminative, which further proves the effectiveness of the proposed SSGA.

3) Execution Time: The execution times of the proposed SSGA approach versus the number of selected bands on all the three considered HSI data sets are shown in Fig. 13 (software implemented in MATLAB with PC workstation [Intel(R) Core (TM) i7-5930K CPUs at 3.50GHz]). As expected, the computation times spent on all three data sets are basically proportional to the number of selected bands. It is due to that the computational complexity of the fitness function and the search process is proportional to the amount of data. Meanwhile, one can see that some exceptions appear in Fig. 13. It may be due to the early stop termination criterion adopted by the SSGA. In greater detail, the execution time of the SSGA has significant differences in the three HSI data sets. The execution time taken on the Indian Pines data set is smaller than those required on the Pavia University and the Pavia Center data sets using the same number of selected bands. It can be attributed to the fact that the time used for the construction of the superpixel maps and computation of fitness values is proportional to the size of data and the number of superpixels, respectively.

Table VII reports the execution times of the proposed SSGA and the compared BS methods when 9 bands have been selected on the three considered HSI data sets. One can observe that the proposed SSGA needs a considerable amount of time to search the optimal subset of bands in an iterative fashion. Nevertheless, the SSGA can select subset of bands that result in more accurate classification performance compared with other methods. Thus, we believe that with the improvement of computing power and the wide use of high-performance computing, the execution time of the proposed SSGA is not a critical issue. Also note that the BS task is done once offline with respect to the classification process.

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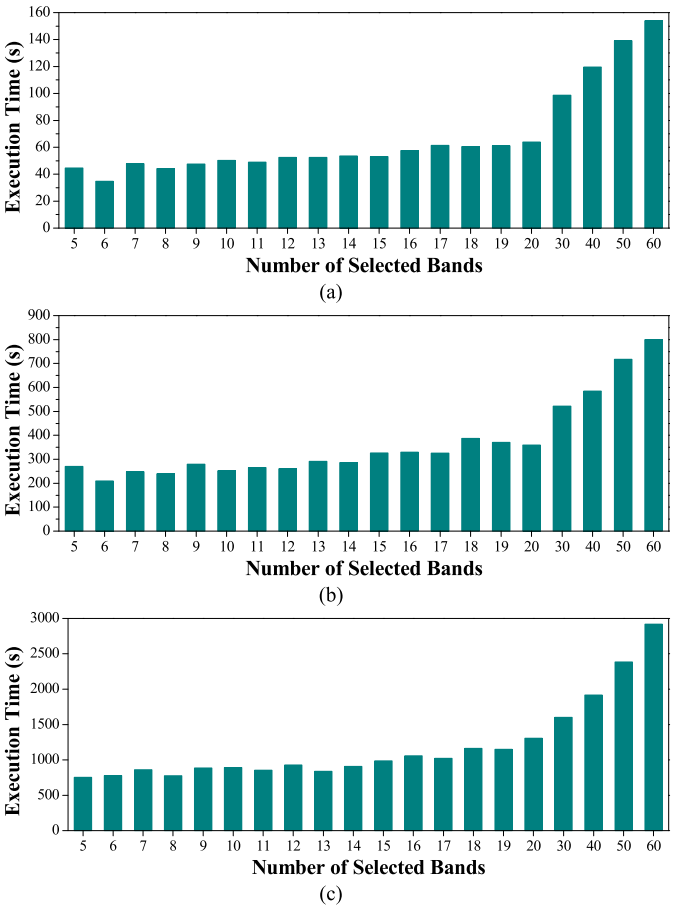


Fig. 13. Execution time (seconds) taken by the proposed SSGA versus the number of selected bands. (a) Indian Pines data set. (b) University of Pavia data set. (c) Pavia Center data set.

IV. Conclusion

In this article, a novel unsupervised BS method based on SSGA (SSGA) has been proposed. In the proposed approach, a superpixel-based fitness function is established by extending the fisher score for considering the spectral features and spatial information of HSI. Meanwhile, new crossover and mutation operations are designed to select a desired number of bands with low redundancy by restraining the search space. To evaluate the effectiveness of the proposed SSGA BS method, a series of experiments have been carried out on three HSI data sets. The effectiveness of the modified crossover and mutation operations used in the proposed SSGA has been confirmed by the experimental comparisons with the GA. In addition, we compared the proposed SSGA with seven state-of-the-art unsupervised BS methods. The experimental results show that the band subset selected by the SSGA has higher discrimination, lower redundancy, and higher stability than those selected by other methods.

In future work, we will focus on exploring more effective genetic operations optimized for BS and designing more flexible fitness functions that can be suitable for various applications.

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